

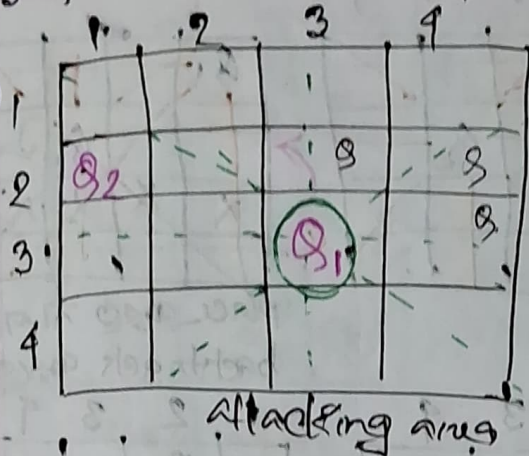
Games

Backtracking

Q₁ Q₂ Q₃ Q₄

N Queens problem. $\rightarrow n: 4, 5, 6, \dots, 7$

4x4
n x n



We have place the in a way
no queen under attack.

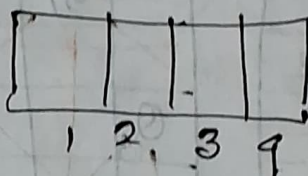
Same

- $\rightarrow$ Row
- $\rightarrow$ Column
- $\rightarrow$ diagonal

rules
avoid attacks area
পাওয়া

Combination

$$16C4$$



① Column

$$C_1 = C_2 \text{ [Not safe]}$$

② Row

$$R_1 = R_2 \text{ [Not safe]}$$

③ Diagonal

$$\text{abs}(C_1 - C_2) = \text{abs}(R_1 - R_2)$$

Q₂, Q₁ এর attacking area
এইভাবে

So, queen প্রত্যেক chessboard এ
অন্যভাবে place করতে হবে যা
attacking mode এর বাইরে থাকবে।

① Q₁ (3, 3) Q₂ (2, 3)

② Q₁ (3, 3) Q₂ (3, 4)

Diagonal $\rightarrow$ ③ Q₁ (3, 3) Q₂ (4, 2)

$$\text{abs}(3 - 4) = \text{abs}(3 - 2)$$

$$1 = 1$$

Backtracking

Q_1, Q_2, Q_3, Q_4

A

	1	2	3	4
1	Q_1			
2				
3				
4				

B

	1	2	3	4
1	Q_1			
2				
3				
4				

C

	1	2	3	4
1	Q_1			
2				
3				
4				

place করে তা নাহলে
backtracks করে

D

	1	2	3	4
1	Q_1			
2				
3				
4				

E

	1	2	3	4
1				
2				
3				
4				

F

	1	2	3	4
1				
2				
3				
4				

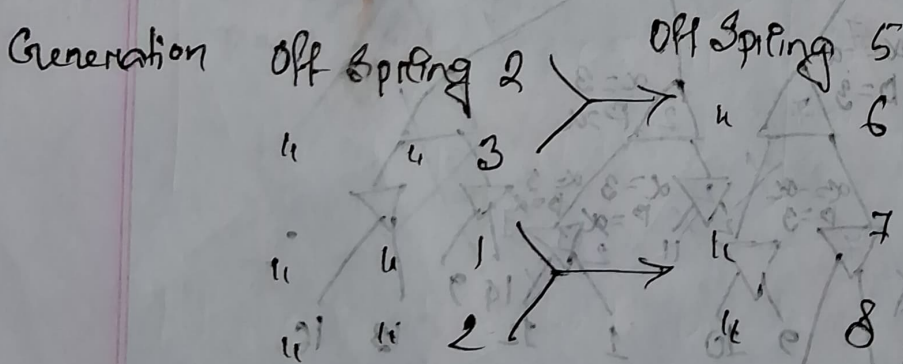
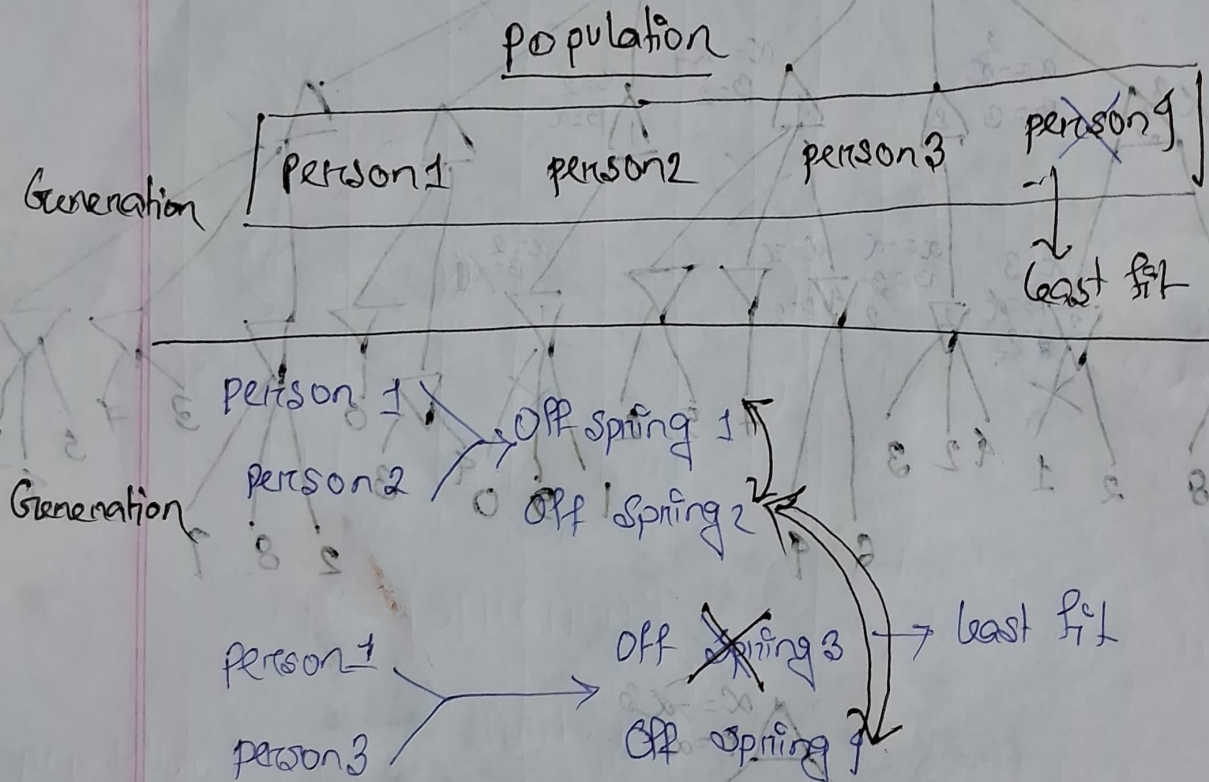
* আমরা backtrack করলে আগের Queen নতুন করে place করতে হবে,

* C তে দেখতে পাচ্ছি row 3 তে কোনো Queen placed হয়নি। তাই বাকি track করে row 2 তে চান্স। and Q_2 কে remaining করে place ২ বসানো।

* D তে দেখা যাচ্ছে row 4 অগুন blocked। তাই backtrack করে করে place ৩ check করতে হবে, but row 3 তে করে place নেই, then অগুন backtrack. এভাবে হবে until we get the best fit.

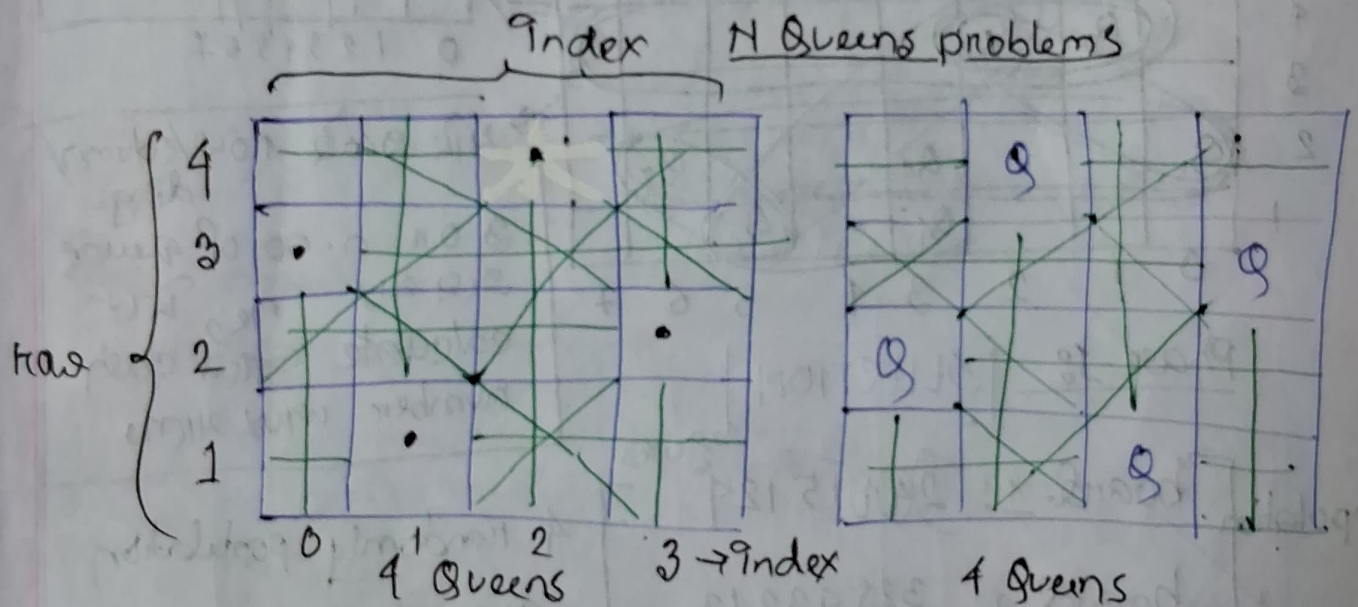
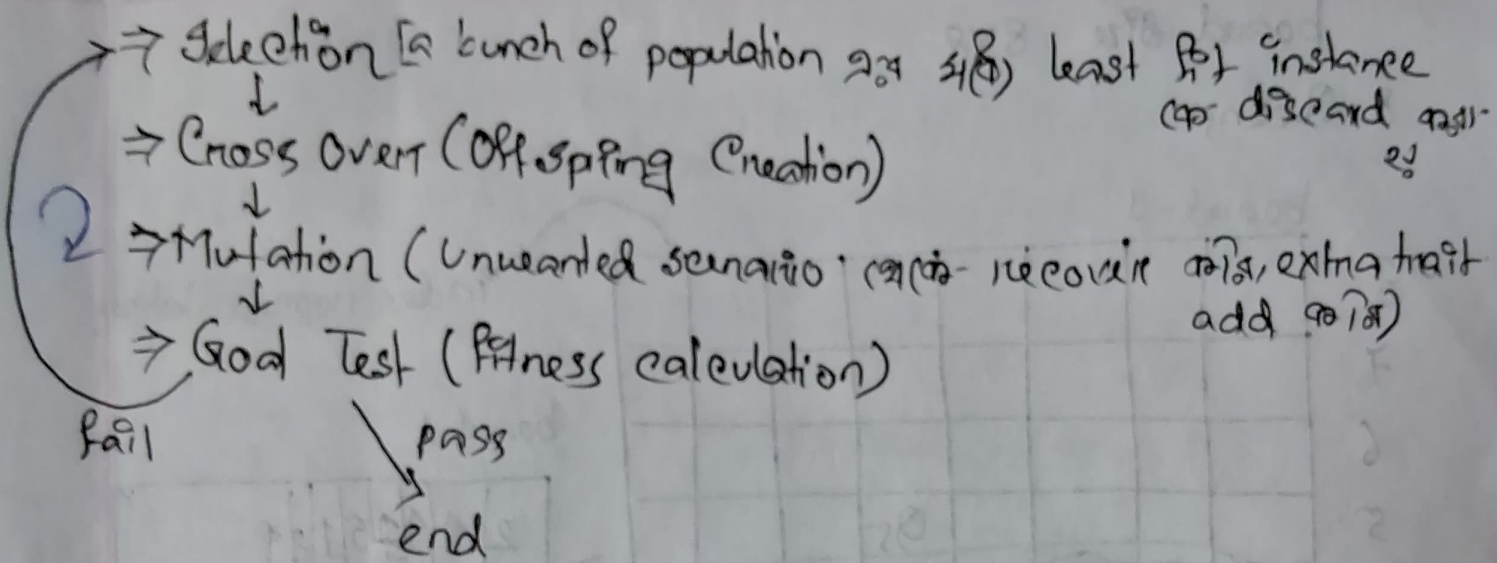
Genetic Algorithm

Evolution theory idea
(See Survival of the fittest)

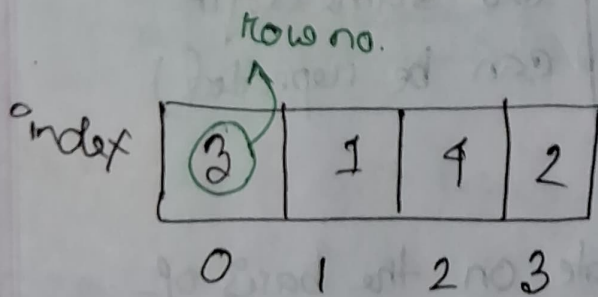


⇒ one generation to another generation, transferring genes (good quality)
we get a perfect goal generation at the end.

Genetic algorithm steps



⇒ 3 1 4 2

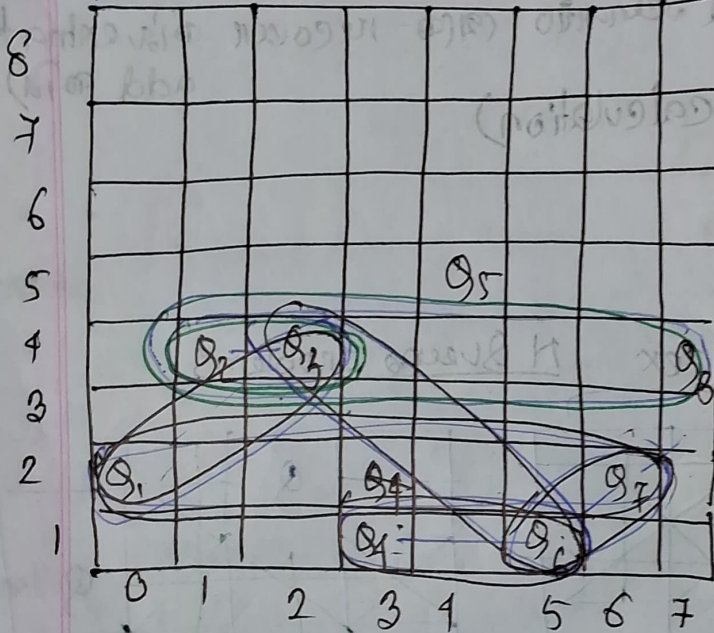


8 Queens Problem

board size 8x8

8 Queens

board-1



Phase-1

SELECTION

genes

population

chromosomes

board-1: 24415124

board 2: 32543213

board 3: 32752411

board 4: 24748552

index representation

board 1

2	4	4	1	5	1	2	4
0	1	2	3	4	5	6	7

4th row/col/diag.

3 on n. no. of queens

2(4) 2(1) 1(1) 1(1)

calculate clash

number (1+1+1+1)

4 randomly population generation

(So same digit can be repeated)

Fitness function will operate on the bases of no. of non-clashing pairs.

no. of non clashing pair $\rightarrow$ fitness maximized

From board 1,

possible combination of pairs

$$8C_2 = 28$$

Clashed pair in board 1

$$1 + 1 + 1 + 1 + 1 + 3C_2$$

$$= 5 + 3$$

$$= 8$$

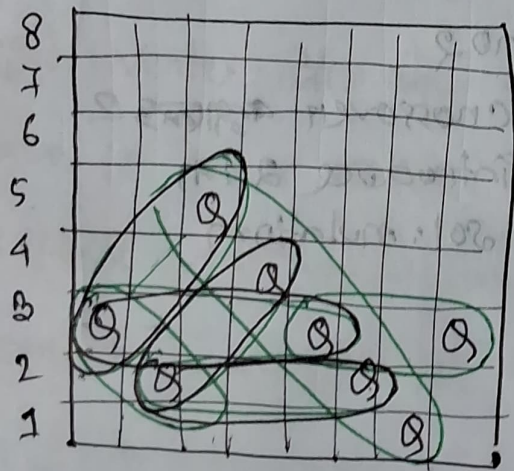
$$\text{Fitness} = 28 - 8$$

$$\downarrow$$

$$= 20$$

no. of
non
clashing
pair

From board - 2



$$1 + 1 + 1 + 1 + 3C_2 + 5C_2$$

$$= 4 + 3 + 10$$

$$= 17$$

Total combination = 28

$$\therefore \text{Fitness} = 28 - 17$$

= 11 (lowest fitness)

From board - 3 $\rightarrow$ fitness = 23

" " 4 $\rightarrow$ " : 24

CROSSOVER

Chromosome - 1: 24415129
" - 2: 32752411

" - 1: 24415129
" - 4: 24748552

24415411
32752129
24415153
24748529

offspring

8x8 board

(1 to 8) (parent) random gene change
 $n \times n \rightarrow (1 \text{ to } n)$

MUTATION:

8

24415411 $\rightarrow$ 28415411

2

X 32752129 $\rightarrow$ 32752129

3

24415152 $\rightarrow$ 24415352

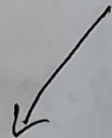
3

24748529 $\rightarrow$ 24738529

now we've got next generation,
 on population,

then $\rightarrow$ fitness calculate

$\rightarrow$ goal test (no of clashes = 0)



$(82) = 28$
 and fitness
 28 too

3313
 3411
 4313
 4143

no 2

crossover 3 2
 introduce 2 (n)
 sol: mutation

random (parent) change

$\rightarrow$ new traits introduce
 2 (n)

$\rightarrow$ parent not a (parent)
 random feature absent
 random child a
 random way mutation
 through offspring

$\rightarrow$ chromosome
 a repetition 2 (n)
 clash 2 (n)

